Replace the Linear Probe: Adapting Feature-Based Model Search to Regression

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ABSTRACT

Model search selects the most promising pretrained model from a candidate pool by ranking candidates with a cheap proxy score instead of fine-tuning each one. Existing systems such as Alsatian and SHiFT target classification. We adapt feature-based model search to regression over tabular data serialized to text. On a pool of 16 text encoders, 14 of them fine-tunable, we evaluate every candidate twice, once with a proxy head on the frozen backbone and once with the backbone fully fine-tuned as reference. We measure the gap as regret with cross-seed error bars against a random-ranking null. Adaptation required three regression-specific changes: replacing the linear probe with a small two-layer perceptron head, stabilizing the fine-tune reference through head warmup and target z-scoring, and treating the quality metric itself as part of the adaptation, since r^2 collapses the entire pool to zero on a long-tailed second dataset whose ordering the models demonstrably learn. The reference ranking is reproducible across seeds (intraclass correlation 0.74), yet the frozen proxy does not reproduce it (rank correlation 0.20). Regret is nevertheless near zero from a shortlist budget of 2 and zero from 3 (≤ 0.001). Only at budget 1 does the proxy fall behind the random null, for one systematic reason: its top pick is a static embedder that cannot be fine-tuned. Filtering the pool for fine-tunability removes that failure and puts regret at budget 1 below the null as well (0.006 vs. 0.018). The classification bottleneck carries over: feature extraction accounts for 73–78% of proxy wall-clock, and the fine-tuning sweep costs 15× the proxy sweep.

1 INTRODUCTION

Public model hubs like Hugging Face (HF) offer over a million pretrained models [25]. Fine-tuning every model to find a well-suited model for a specific task is infeasible. Therefore, *model search systems* such as Alsatian [25] and SHiFT [20] rank candidate models by performance using a cheap *proxy*, typically a linear probe or k-nearest neighbor (kNN) trained on the frozen-backbone's features to signal how well a model performs. As a result, only the best ranked models need fine-tuning. Model search plays a vital role in Just-In-Time Model Replacement (JITR). JITR aims to select and train a cheap surrogate model to stand in for a costly overqualified model performing a recurrent task, as envisioned in Poodle [26].

These model search systems were designed and evaluated solely for classification tasks. Model search systems for regression remain unexplored, particularly in combination with text encoders. Many prediction tasks are regression on tabular data (e.g. price prediction of property, or rating of some good). This data can be serialized to text and passed to the same candidates of encoder models. Whether previous work on model search carries over to regression is yet to be shown. This report concerns two research questions:

- **RQ1 (Adaptation).** How can feature-based model search be adapted to text regression, and is a frozen-backbone proxy ranking a reliable substitute for fine-tuning every candidate?
- **RQ2 (Bottlenecks).** Alsatian identified feature extraction (model loading, data preparation, and inference) as the dominant cost of model search for classification tasks. Does this also hold true for regression tasks, and how do the costs of the proxies compare to full fine-tuning?

We will measure results regarding *RQ1* using *regret*, a metric from SHiFT [20] (Section 3.4). Regret quantifies the gap in quality between selecting top-ranked candidates and the actual top candidate. We answer both our research questions on a candidate pool of 16 pretrained text encoders and on two tabular regression datasets. Each candidate is evaluated twice: once with a cheap proxy attached to its frozen-backbone, and once with the backbone fully fine-tuned as reference. All our metrics for quality are standard regression metrics against known labels. Motivated by JITR's scarce-data setting, we answer the small-data-budget question, reported separately: does the proxy ranking hold up when a new backbone must be judged on roughly 2,000 rows? This report contains the following contributions:

- We show that adapting feature-based model search to regression requires three changes: a small multi-layer perceptron (MLP) head instead of a linear probe; reference stabilization through LP-FT warmup [8] and target z-scoring; and a per-task quality metric, since r^2 is near zero across the pool on a long-tailed target, even though the models correctly rank candidates by price.
- We measure proxy faithfulness with seed error bars against a random-ranking null (Section 3.4): the reference ranking is reproducible (intraclass correlation coefficient (ICC) = 0.74) yet the proxy cannot rank it ($\rho = 0.20$); its regret is near zero from budget 2 and zero from 3, but at budget 1 it is worse than random because it nominates a candidate that cannot be fine-tuned, hinting at the need for a fine-tunability filter.
- We confirm that the same bottleneck identified by Alsatian [25] for classification still persists for regression: feature extraction accounts for 73–78% of proxy wall-clock

(Section 5.2), while the fine-tuning sweep costs 15× the proxy sweep (Section 4.9).
- We show that on the primary dataset (Section 3.1) the proxy is consistent even when capping the rows at 2,000 ($\rho = 0.88$ against the full-data proxy ranking (Section 4.8)) without reintroducing training instability.

The full pipeline, experiment configurations, and analysis code are available at https://github.com/EnnioEnnio/MLSys.

Section 2 reviews related work, Sections 3 and 4 present the methodology and experiments, and Sections 5 and 6 answer the research questions and outline future work, including an exploratory transfer pilot on text summarization.

2 RELATED WORK

Model search systems. SHiFT [20] is a model search engine for transfer learning: it selects a model from a pool of candidates for a given target task by ranking candidates via cheap proxies. It formalizes search quality as *regret* — the fine-tuned accuracy lost by trusting a proxy ranking instead of fine-tuning every candidate. Alsatian [25] identifies feature extraction (model loading, data preparation, and inference) as the dominant bottleneck in classification-task model search. Sommelier [6] indexes models by resource-accuracy tradeoffs, meaning resource consumption relative to prediction performance, rather than scoring candidates for a new target task, which is an orthogonal objective to ours. We adopt SHiFT's regret metric (Section 3.4) and Alsatian's cost-decomposition methodology (Section 3.5); both systems were designed and evaluated for classification, leaving open whether the bottleneck transfers to regression.

Proxy strategies. Renggli et al. [19] compared proxy strategies and concluded that no single strategy dominates across tasks. One approach uses closed-form transferability estimators as a proxy for model performance. Estimators such as H-score [1] and LEEP [14] target classification, while LogME [29] extends to regression, but is only validated on continuous vision targets. Even estimators tailored specifically for regression, Linear MSE and Label MSE [13], are solely evaluated on vision regression benchmarks. Instead of an estimator, we opted for a small trainable regression head (Section 4.4). None of this prior work combines regression with text encoders.

Two-phase fine-tuning. Kumar et al. [8] show that fine-tuning with a randomly initialized head distorts pretrained features. Their proposed approach, linear probing then fine-tuning (LP-FT), trains only the head on the frozen backbone, then unfreezes and fine-tunes. ULMFiT [7] introduces gradual unfreezing of the backbone, a finer-grained alternative to LP-FT [8], exploiting the fact that higher layers are more task-specific than lower ones, as shown by Yosinski et al. [28]. In our case, a randomized head corrupts the fine-tuned reference ranking that regret is measured against. This shows that reference stabilization is part of the search problem.

3 METHODOLOGY

We adapt model search to text regression and use it to analyze our research questions. Given a pool M of 16 pretrained text encoders (Section 3.6) and a target dataset D, the pipeline scores every candidate twice: a cheap *frozen proxy pass* that produces the search ranking, and an expensive *fine-tune reference pass* that produces the ground truth against which the ranking is validated. The fine-tuning recipe is designed to train stably and to produce consistent results before any regret is measured (Section 3.3). We quantify the quality gap as regret with cross-seed error bars against a random-ranking null (Section 3.4). Every run is decomposed into timed substeps used for the bottleneck analysis (Section 3.5).

3.1 Task Setup

Each benchmark scenario is a regression dataset with a numeric target. Tabular rows are rendered to text through a per-dataset template, turning the tabular regression into a text-regression task that any text encoder can consume.

The primary dataset is `wine_reviews` [24]. For this dataset, every row is cast to a string of the form `description: {description}, variety: {variety}, country: {country}, price: {price} USD`. The regression target is the review score in points on the WineEnthusiast theoretical 0–100 scale (train mean 88.15); only points 80–100 are populated in practice. A mean absolute error (MAE) of 1.4 therefore means predictions land within about 1.4 points on a 20-point effective range. We use the fixed HF splits: 196,630 train, 28,090 validation, and 56,181 test rows. The validation split is used to drive early stopping and all reported metrics are computed on the test split.

The secondary dataset is `usa_real_estate` [21] with the long-tailed `price` column as regression target. The dataset is a single physical split, which we shuffle with a fixed seed and slice to exactly the sizes of the primary dataset's splits, to enable comparability.

Both datasets additionally have a tiny variant with 2,000 train and 1,000 validation rows, which mimics a data budget of the model-replacement scenario in Section 4.8. The tiny wine variant uses the same test split as the full run, making r^2 values directly comparable. The tiny housing test split is a same-sized slice of the shuffled pool.

3.2 Proxy and Reference Passes

For every candidate encoder in the pool, the pipeline runs two passes.

Frozen proxy pass. A fresh, seed-initialized, fully connected regression head is attached to the frozen backbone. All rows are embedded once, only the head is trained against the ground-truth targets (AdamW [11] with mean squared error (MSE) loss and early stopping after three epochs). The resulting model is scored on the test split. This proxy is the regression analogue of the linear-probe and kNN proxies discussed for classification in Alsatian and SHiFT respectively [20, 25].

Fine-tune reference pass. First, the backbone is unfrozen, then the backbone and head are trained jointly, and finally evaluated on the same test split. This pass provides (i) the reference signal $t(m, D), m \in M$ that the proxy ranking is validated against and (ii) the per-model fine-tuning cost used to evaluate RQ2 (Bottlenecks).

The `full_eval` strategy runs both passes over the whole pool and computes the regret-vs-budget curve.

Head architectures. A head is either a single linear layer, the fully connected head (FCH) configuration acting as a linear probe, or a 2-layer MLP, with input dimension $d \in \mathbb{N}$, $\text{Linear}(d, h) \to \text{ReLU} \to \text{Linear}(h, 1)$, with hidden width $h \in \{128, 256, 512\}$, named `MLP_128`, `MLP_256`, and `MLP_512`.

Table 1: Training configuration for the frozen proxy pass (head only) and the fine-tune reference pass (joint).

	frozen head	fine-tuning (joint)
epochs (max)	30	10
batch size	256	64
optimizer	AdamW	AdamW (two param. groups)
learning rate	10^{-3}	backbone $2 \cdot 10^{-5}$, head 10^{-3}
weight decay	10^{-4}	10^{-4}
early stopping	patience 3, min-delta 10^{-3}, best-state restore	
loss	MSE	MSE
head warmup	—	2 epochs head-only (LP-FT)
target z-scoring	on (train stats)	on (train stats)

Metrics. We record MSE, MAE, r^2, and the Spearman correlation of each model's predictions against the targets. r^2 is the primary metric; regret is computed on r^2 only. This per-model prediction Spearman is distinct from the *rank agreement* between the proxy ranking and the reference ranking of the pool. Both metrics are covered in detail in Section 4.

3.3 Training Configuration

Table 1 lists the training configuration for both passes. All backbone models share the same training configuration. This is a deliberate fairness constraint, since the candidate encoders differ in embedding dimension, tokenization, pooling, and input prefixes.

The final configuration we used for all regret measurements is: `MLP_256` head in both passes, 2-epoch head warmup, target z-scoring, and ReLU activation. We tested gradient clipping and learning-rate decay and found both to be redundant (Section 4.3). The configuration was fixed before the regret measurement and selected as the most stable configuration with zero fine-tune divergences ($r^2 < 0$). The reference serves as a measuring device to validate the proxy ranking, rather than as the optimal result of the fine-tuning.

Targets are z-scored on train-split statistics. Models train in standardized space and predictions are mapped back before scoring. Therefore, MSE and MAE stay in original units, while r^2 and Spearman are unaffected by construction, since both are invariant to affine transformations of the target. Z-scoring stabilizes the reference pass. On raw 80–100 targets, and with 2-epoch warmup already in place, the ELECTRA-style encoders of the pool (`deberta-v3-base`, `deberta-v3-small`, `electra-base-discriminator`) still diverge or land far below the rest of the pool under fine-tuning. The 2×2 experiment we ran after the head sweep isolates target z-scoring as the change that removes all of them (Section 4.3). Head warmup addresses the complementary fine-tuning failure mode analyzed by LP-FT [8]. Feature-distorting gradients backpropagate from a freshly initialized head into the backbone during the first joint steps. To overcome this, we begin to train only the head for N epochs against the frozen backbone before unfreezing. The final configuration uses $N = 2$ (quantified in a controlled ablation in Section 4.2).

3.4 Regret and the Random-Ranking Null

We quantify the faithfulness of the proxy ranking with the regret metric of SHiFT [20] with a small adjustment. Due to the fact that we only fine-tune once per model, the expected values convert to point estimations, which is why we removed them from our formula. Let $t(m, D)$ be the test r^2 of model m *after fine-tuning* on D, and let $S_B \subseteq M$ be the top-B shortlist by frozen-proxy r^2. Then our regret is defined as:

$$\text{regret}(B) = \max_{m \in M} t(m, D) \; - \; \max_{s \in S_B} t(s, D). \tag{1}$$

Regret is non-negative, lower is better, and zero means the list already contained a model as good as the global optimum. We report absolute regret as a full regret-vs-budget curve, plus *budget-to-zero*, the smallest B with zero regret, and the Spearman ρ between the proxy and reference rankings.

The headline regret numbers (Section 4.6) come from five seed repeats of the full protocol under the final configuration and are reported as mean with min–max band.

The random-ranking null. A low regret value alone is ambiguous. It can mean the proxy ranks well, or that the task forgives any shortlist. We separate the two with a Monte-Carlo null: per seed, 10,000 uniformly random orderings of the pool are drawn over the 16 measured fine-tune scores, computing regret(B) exactly as for the real proxy ranking, and the 50,000 samples pooled over the five seeds are reported per budget as mean with 5–95% percentile band. The proxy earns its keep only where its regret curve sits clearly below the null band.

3.5 Cost Instrumentation

Following the cost-decomposition methodology of Alsatian [25], every candidate run is split into five timed substeps. These substeps are model loading (`prepare_model_s`), data materialization and rendering (`prepare_data_s`), embedding inference (`inference_s`), training (`train_head_s`), and test evaluation (`eval_s`), with peak graphics processing unit (GPU) memory and epochs run recorded per candidate. The attribution differs structurally between the frozen-proxy- and fine-tuning-passes. The frozen pass separates backbone inference from head training, whereas fine-tuning fuses inference into the training. Therefore, its `inference_s` is zero and `train_head_s` covers both, inference and training including warmup. This is important for comparing the two cost profiles.

3.6 Model Pool

The pool contains 16 encoders across three adapter families (full list in Table 8 in the appendix): `sentence_transformers` models with built-in pooling, plain `transformers_encoder` models with mean or CLS pooling over token states, and `model2vec` [27] static embedders. The static embedders have no transformer forward pass and cannot be fine-tuned. Under the reference pass they fall back to their frozen score and we flag them as skipped. This heterogeneity turns out to matter for the search itself. Section 4.6 shows the proxy's worst failure is crowning a candidate that cannot be fine-tuned at all.

3.7 Execution Infrastructure

Runs execute on the HPI Slurm cluster as job arrays, one array task per candidate model, so a pool sweep parallelizes across nodes and a single crashing model cannot poison the sweep. A dependent consolidation job merges the per-task fragments, recomputes the regret artifacts, and exports per-run frozen, fine-tune, and regret CSVs. An analysis module turns an experiment folder of such CSVs into tables and plots. Two side-scripts produce the seed-statistics and cross-dataset artifacts of Sections 4.6–4.8. Moreover, the search can also be executed as one single job via `sbatch`. The pipeline, the job-array and consolidation scripts, the analysis module, and the two side-scripts are available at https://github.com/EnnioEnnio/MLSys.

Each job runs on a single NVIDIA A100 GPU with 64 GB host RAM. All wall-clock and peak-GPU-memory numbers in this report are measured on this A100 configuration. The largest observed memory consumption, about 28 GB for `mxbai-embed-large-v1` under fine-tuning, fits within a single A100.

4 RESULTS AND EVALUATION

Sections 4.1–4.5 are design-space experiments that shape and probe the measurement configuration; they inform design decisions only. Section 4.6 is the RQ1 measurement: five seed repeats of the full protocol against the random-ranking null. Sections 4.7 and 4.8 cover the secondary dataset and the small-budget replacement scenario, Section 4.9 the RQ2 costs. All experiments run the full 16-model pool; the verdicts follow in Section 5.

4.1 Proxy Repeatability

The frozen proxy trains a randomly initialized head per candidate, so we first check whether a single head fit is a stable ranking signal. On the frozen pass with the linear FCH, we compare one head fit per model against a three-seed ensemble over the same cached embeddings, with predictions averaged before scoring. The two proxy rankings are nearly identical ($\rho = 0.991$ across the 16 models). The top model (`e5-base-v2`) and the top-3 rankings match exactly, and per-model differences are small (maximum $|\Delta r^2| = 0.012$, mean 0.002; full table in Appendix A.2). Head-training cost meanwhile scales linearly with the repeat count (860 s to 2589 s), so all subsequent experiments use a single head fit per candidate. This covers proxy-side variance only; Section 4.6 answers the reference side.

4.2 Head Warmup

Regret validates the proxy ranking against fine-tuned ground truth, so diverging fine-tunes corrupt the measuring procedure itself. Naive joint fine-tuning is not a valid reference: without stabilization, 3 of 16 backbones diverge. This section quantifies the problem and the LP-FT warmup fix; Section 4.3 completes the stabilization.

Epoch sweep. The experiment sweeps the warmup budget over $\{0, 1, 2, 4, 8\}$ epochs (fine-tune pass only) for the full model pool with the `MLP_256` head (the width is justified by the head sweep in Section 4.4) and 10 joint epochs. Each model is fine-tuned once per budget, so individual r^2 deltas are noisy. We focus on the divergence count and the pool mean as robust signals. Table 2 reports the curve.

Table 2: Warmup-epoch sweep (fine-tune pass, `MLP_256`, 16 models, one run per model and arm; pre-z-scoring protocol). The 0- and 2-epoch arms are the two-point ablation.

warmup epochs	0	1	2	4	8
diverged ($r^2 < 0$)	3	0	1	0	1
mean fine-tune r^2	0.553	0.762	0.736	0.787	0.740
median fine-tune r^2	0.798	0.859	0.862	0.866	0.876
mean r^2, 13 stable models	0.758	0.807	0.805	0.825	0.820

Table 3: Stabilizer 2×2 on top of 2-epoch warmup (`MLP_256`, 16 models, one run per cell). The trio columns give fine-tune r^2 for the three unstable models.

cell	diverged	`deb-base`	`deb-small`	`electra`
warmup-2 only	1	−0.49	0.62	0.43
+ grad clipping	1	−0.51	0.59	0.41
+ target z-scoring	**0**	**0.89**	**0.90**	**0.91**
+ both	**0**	no better than z-scoring alone		

We choose a 2-epoch head warmup, a balance between the added warmup epochs and the stability they buy.

Any warmup of at least one epoch clears the cold-start divergences (3 to 0), rescuing `deberta-v3-small` (−0.258 → +0.771 r^2 in the 2-epoch configuration) and `roberta-base` (−0.118 → +0.788). The divergence count of 1 at 2 epochs versus 0 at 1 and 4 is the same single chronic model (`deberta-v3-base`) which does not follow a monotonic trend but is unstable. All diverged runs keep a prediction Spearman of 0.86–0.93.

The best model (`modernbert-embed-base`) reaches $r^2 \approx 0.91$ with or without warmup, and among the 13 models already stable without it, warmup buys only a small bump that peaks at 4 epochs (+0.067 over cold start) before plateauing. The budget-to-zero and rank-ρ differences between the arms are artifacts of which models diverged and carry no signal. The warmup-resistant `deberta-v3-base` suggests a model-architecture-side cause, addressed next.

4.3 Clipping and Target Standardization

To resolve the warmup-resistant divergence, a 2×2 experiment crosses gradient clipping (max norm 1000) with target z-scoring, on top of 2-epoch warmup (full pool, `MLP_256`, one run per cell). Following the other configuration shaping experiments, arms are judged on divergence counts and the unstable models, not on mean r^2. Table 3 reports both, with the unstable trio—`deberta-v3-base`, `deberta-v3-small`, and `electra-base-discriminator`—broken out.

Raw 80–100 targets produce substantial initial loss and gradient spikes (`deberta-v3-base`: mean norm around 200, spikes to around 30,000). Standardizing the target removes the spikes at the source: all divergences disappear and the unstable `electra`-family trio reaches the same r^2 range as the rest of the pool.

This fixes the final configuration of Section 3.3: warmup-2 plus target z-scoring, `MLP_256`, ReLU, with clipping and learning-rate decay tested and found redundant. A frozen-only activation sweep (ReLU, GeLU, tanh, SiLU at `MLP_256`) kept ReLU in place; tanh had

Table 4: Frozen-proxy r^2 distribution and early-stopping behavior per head (16 models).

head	mean	std	min	$n_{r^2<0}$	avg. ep.	at cap
FCH	0.461	0.305	−0.248	2	29.1	12/16
MLP_128	0.646	0.051	0.490	0	15.6	0
MLP_256	0.657	0.057	0.478	0	14.7	0
MLP_512	0.669	0.069	0.430	0	13.8	0

the best mean proxy r^2 (0.716 vs. 0.655), but the sweep's notable outcome is that the proxy *ranking* moves with the activation: pairwise rank agreement between activations is only $\rho = 0.37$–0.69, and the proxy top-1 differs per activation. The proxy head is not a detail; its design choices reshape the ranking.

4.4 Proxy Head Type and Width

With warmup fixed at 2 epochs (pre-z-scoring), we sweep the head architecture over FCH, MLP_128, MLP_256, and MLP_512 in a full evaluation of the pool (one fine-tune run per model and head). This experiment justifies deviating from the linear-probe default of the classification literature; the model × head r^2 heatmaps and full per-model tables are in Appendix A. We choose MLP_256 in both passes: one hidden layer is what the proxy needs, wider heads add nothing on either side, and 256 keeps the width of the preceding experiments.

Reference side. The best achievable r^2 is about 0.90 at every width. With the fine-tuned backbone, head capacity becomes less important for the models that converge, and the rising per-head mean (0.586 under FCH to 0.737 at MLP_512, Table 10 in the appendix) is driven by the non-finetunable potion models, whose features are not linearly decodable. They require nonlinearities to converge. deberta-v3-base fine-tunes to $r^2 \leq 0$ at every width and deberta-v3-small collapses at MLP_512 (0.137, from 0.771 at MLP_256). Both reflect the pre-z-scoring instability at one run per cell, orthogonal to head capacity. Together with our LP-FT findings, this motivated the introduction of target normalization.

Proxy side. Table 4 shows the frozen-proxy distribution per head. The linear probe fails on three counts: it scores negative r^2 for two encoders (all-MiniLM-L6-v2 at −0.25, potion-base-8M at −0.11), its score spread is six times that of any MLP, and it is the slowest to train, with 12 of 16 models hitting the 30-epoch early-stopping cap without converging. One hidden layer (MLP_128) collapses the spread to $\sigma \approx 0.05$, removes every negative score, and roughly halves the training epochs; widths 256 and 512 add little. The models that gain most from widening are exactly the ones a linear probe cannot read from frozen embeddings: all-MiniLM-L6-v2 (+0.86), potion-base-8M (+0.82), potion-base-32M (+0.65), and all-mpnet-base-v2 (+0.51). Models already scoring at least 0.62 under FCH gain less than 0.10 from a wider head. The head choice also reshuffles the proxy ranking itself: rank agreement between the FCH and MLP_512 proxies is only $\rho = 0.17$, and potion-base-32M sits at rank 14 under the linear probe but rank 1 under MLP_512 (rank 3 under MLP_256).

Ranking and regret, single-run. The proxy-vs-reference rank correlation is near zero at every width (ρ between −0.18 and +0.12). The per-head regret values are single-run point estimates and carry no information individually: regret@1 spans $[0.00, 0.13]$ across seeds (Section 4.6). One value is signal: at MLP_512 the regret@1 jumps to 0.176 because widening the head flips the proxy top-1 from e5-base-v2 to potion-base-32M, the non-finetunable static embedder whose reference is capped at its frozen score of 0.725; the gap is exactly 0.901 − 0.725. Section 4.6 shows this trap firing systematically under the final recipe.

4.5 Different Activation Functions

Section 4.3's frozen-only sweep showed that activation choice reshapes the proxy ranking itself, not just its mean r^2. However, a frozen-only comparison cannot say whether tanh's ranking is better, only that it is different. To answer that, we ran a full evaluation keeping the final recipe fixed and swapping only the activation to tanh.

Frozen-pass mean r^2 was slightly lower under tanh than under ReLU, compared against an equivalent ReLU full-evaluation run. But the Spearman rank correlation between the fine-tuned reference and the frozen proxy was higher under tanh (0.385 to 0.662, Table 15), meaning the proxy ranks candidates more in line with the reference than it does under ReLU. Both values are single runs. The final correlation we report is the five-seed $\rho = 0.20$ $[0.07, 0.30]$ of Section 4.6, and the ReLU run here sits at 0.385, outside that band, so the tanh gap is a lead for a seed-repeated rerun (Section 6), not a measured effect.

On the fine-tune side, tanh did not improve the reference on average across the full 16-model pool, landing slightly below ReLU. This gap is driven almost entirely by the two non-finetunable model2vec backbones, whose frozen fit is significantly worse under tanh. Restricted to the 14 backbones that are actually fine-tunable, the two activations are effectively tied, and the top-performing backbone's fine-tuned r^2 is unchanged. So tanh does not degrade real fine-tuning outcomes, it mainly improves how well the cheap proxy ranking tracks them.

4.6 Regret, Noise Floor, and Random Null

All results below use ReLU; Section 4.5 flags a single-run tanh result as a possible exception, unconfirmed across seeds. The headline RQ1 measurement is five complete full_eval repeats (both passes, 16 models each) under the final configuration on wine_reviews. Under this recipe every trainable model lands at fine-tune r^2 0.87–0.92, which admits two readings: either the frozen proxy genuinely cannot rank fine-tuned quality, or the reference ranking itself is not reproducible across seeds, in which case regret would be noise measured against noise. The following seed statistics provide insights.

The recipe holds. Across all 5×14 trainable fine-tunes there is not a single divergence. The zero-divergence recipe, selected on one run, survives seed variation.

The reference is reproducible. Table 5 gives the noise statistics. A single fine-tune's r^2 moves by $\sigma_{\text{within}} \approx 0.008$ across seeds (worst model albert-base-v2 at std 0.026), while the trainable band of seed-mean scores is 0.046 wide, about six times that noise floor; the ICC of 0.74 means the trainable models remain statistically

Table 5: Noise statistics over 5 seeds: within-model std, between-model std of seed means, their ratio, ICC, and the band of seed-mean r^2.

pass	σ_{within}	σ_{between}	ratio	ICC	band
frozen	0.011	0.060	5.4	0.97	0.533–0.793
fine-tune (16)	0.007	0.047	6.3	0.98	0.747–0.918
fine-tune (14 train.)	0.008	0.013	1.7	0.74	0.872–0.918

Table 6: Observed regret (mean [min–max] over 5 seeds) versus the random-ranking null (mean [5–95%] over 50,000 samples), for the full pool and with the two non-finetunable candidates removed.

B		observed	random null
1	full pool	0.100 [0.000, 0.132]	0.034 [0.000, 0.170]
2		0.006 [0.000, 0.017]	0.013 [0.000, 0.033]
3		0.000 [0.000, 0.001]	0.009 [0.000, 0.021]
1	trainable only	0.006 [0.000, 0.017]	0.018 [0.000, 0.045]
2		0.000 [0.000, 0.001]	0.010 [0.000, 0.024]

distinguishable inside the band. The rankings are stable too: fine-tune rankings from different seeds agree at mean pairwise $\rho = 0.93$ (0.90 trainable-only), frozen rankings at $\rho = 0.90$. One tie sits at the very top: the fine-tune winner flips between `modernbert-embed-base` (3 of 5 seeds) and `modernbert-base` (2 of 5), whose seed means differ by 0.001. Appendix A.5 gives the per-model spread and the per-model seed table.

The proxy cannot rank it. Frozen-vs-fine-tune rank agreement is $\rho = 0.20$ $[0.07, 0.30]$ across seeds. Both rankings are individually stable; they stably disagree. Spearman between proxy ranking and reference sits at $\rho \approx 0.2$: `modernbert-base` sits at frozen rank 10–14 while at fine-tune rank 1–3, and `sentence-t5-base` sits at frozen rank 5–11 while at fine-tune rank 13–14.

Regret against the random-ranking null. Table 6 and Figure 1 contain the final regret numbers. At $B = 1$ the observed regret of 0.100 exceeds the null mean of 0.034, and the failure is systematic: in 4 of 5 seeds the frozen proxy's top pick is `potion-base-32M`, the static `model2vec` embedder that cannot be fine-tuned, so its reference stays at its frozen score of ≈ 0.79 against the ≈ 0.92 optimum. This is the same trap that surfaced at `MLP_512` in the head sweep (Section 4.4), now firing in 4 of 5 seeds. The proxy ranks frozen quality, and the best frozen model is precisely the one that gains nothing from fine-tuning. Removing the two non-finetunable candidates (static `model2vec` embedders) flips the picture: regret@1 drops to 0.006 against a null of 0.018, so the entire regret@1 spike is attributable to the non-finetunable top pick, not to ranking error among real fine-tunes. From $B = 2$ the proxy's shortlist always contains a near-optimal model (mean regret 0.006 at $B = 2$, worst seed 0.017), while the null converges far slower: its mean regret only falls below 0.001 around $B \approx 11$ and its 95th percentile is still about 0.01 at $B = 9$. Single-run regret@1 spans $[0.000, 0.132]$ across seeds, which is why the single-run regret values of the design-space experiments carry essentially no information individually.

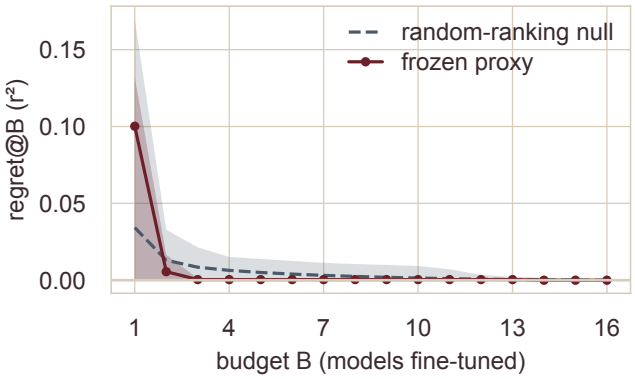


Figure 1: Observed regret band (5 seeds) versus the random-ranking null band per budget B. The proxy sits below the null everywhere except $B = 1$, where its top pick is a non-finetunable candidate in 4 of 5 seeds.

4.7 Secondary Dataset: usa_real_estate

The secondary dataset's job is to find a task where the pool actually separates, so the proxy has something to get wrong. Under the final configuration, with splits sized exactly like our primary dataset and early stopping disabled to rule out under-training, that job is unmet, and the reason is itself a finding. Frozen r^2 spans $[-0.000, 0.023]$ and fine-tune r^2 spans $[-0.000, 0.031]$: the entire pool, both passes, is compressed into a 0.03-wide band at zero, and proxy-vs-reference rank agreement is $\rho = -0.04$; ordering inside a band this narrow is noise.

Interestingly the models learn the task's ordering. Per-model prediction-vs-target Spearman reaches 0.78 after fine-tuning (mean 0.65 over trainable models) while r^2 stays near 0.03; Figure 2 contrasts this with the primary dataset, where the per-model Spearman tracks r^2. The failure is in the metric-target pairing, not in the encoders: house prices are extremely long-tailed, squared error is dominated by tail listings, and the variance denominator of r^2 makes every model look like the mean predictor. Target z-scoring cannot help, it fixes optimization problems (the primary dataset divergences) but not metric problems. A fix must be nonlinear: a log-price target, outlier clipping, or a rank-based quality metric (Section 6).

4.8 Model Replacement Under a 2,000-Row Budget

This section reports the secondary replacement question and is not part of the RQ1 regret numbers. At replacement time a new backbone must be judged quickly on the available data, so both datasets get a run with 2,000 train and 1,000 validation rows under the final recipe, scored on a full-size test split. The focus is rank survival, not absolute r^2. Table 7 summarizes.

On our primary dataset, the proxy ranking survives the small budget: a frozen proxy fitted on 2,000 rows reproduces the full-data proxy ranking at $\rho = 0.88$, with zero divergences and a tiny-run regret@1 of 0.012. Absolute scores drop as expected on the identical test split (frozen mean 0.73 to 0.58, fine-tune 0.88 to 0.68). What



Figure 2: Metric-fragility diagnosis on `usa_real_estate`: per-model prediction-vs-target Spearman against r^2, contrasted with primary. The models rank prices well while r^2 collapses to zero.

Table 7: Rank survival under the 2k-row budget: Spearman ρ between 16-model rankings. The last row is the full-data proxy-vs-reference agreement for context.

comparison	wine	housing
tiny frozen vs. full frozen	**0.88**	0.34
tiny fine-tune vs. full fine-tune	0.62	0.56
tiny frozen vs. tiny fine-tune	0.49	0.39
full frozen vs. full fine-tune (context)	0.20	−0.04

the tiny proxy still cannot do is rank fine-tuned quality: the 2k-row frozen ranking agrees with the full-data fine-tune ranking at $\rho = 0.14$, the same proxy-vs-reference comparison that yields 0.20 at full scale.

On the secondary dataset the full-data frozen ranking spans only 0.000–0.023, so the tiny-vs-full agreement of 0.34 ranks noise, and the tiny run scoring higher than the full one (fine-tune max 0.26 vs. 0.03) is a long-tail sampling artifact, not a small-data win: the 2,000-row train subset holds fewer extreme prices. The tiny housing numbers carry no evidence about model replacement.

4.9 Cost Breakdown

Cost numbers are 5-seed means from the final-configuration run on our primary dataset (Section 4.6). All numbers are measured on the A100 configuration of Section 3.7. This section answers RQ2.

Proxy vs. fine-tuning. A full-pool proxy sweep costs about 0.97 h; the fine-tune reference sweep costs about 14.4 h, a 15× gap at pool level and 6–23× per trainable model (worst: `mxbai-embed-large-v1` at 23×). The two `model2vec` candidates cost the same in both passes; they are their frozen run. Peak GPU memory is at most 2.8 GB for the proxy against up to 27.6 GB when fine-tuning, up to 10× per model. Recall from Section 3.5 that fine-tuning fuses inference into the joint loop, so the two cost profiles attribute time differently.

Proxy Ranking. Figure 3 breaks the frozen pass into its sub-steps. Backbone inference dominates at 47–51% of wall-clock, with data

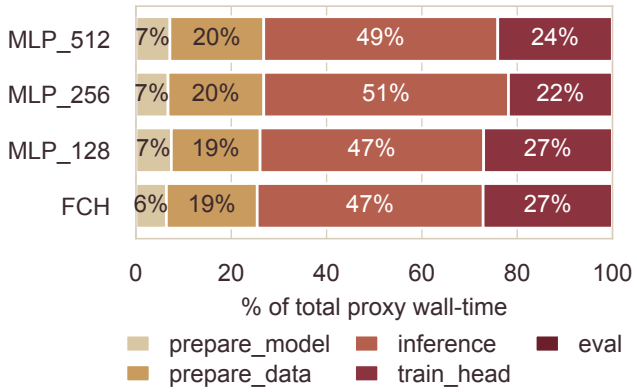


Figure 3: Frozen-proxy wall-clock share per sub-step and head. Feature extraction (prepare model, prepare data, inference) accounts for 73–78%.

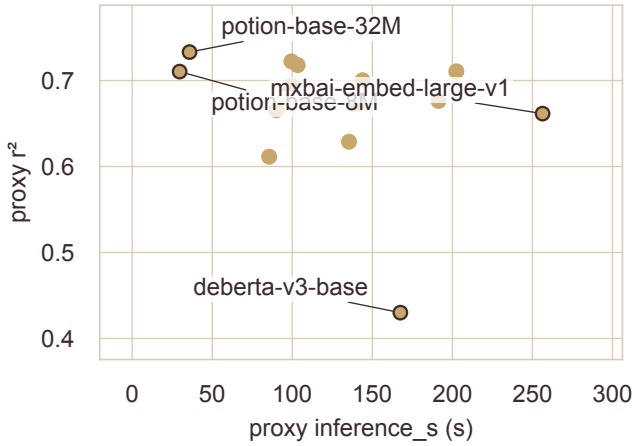


Figure 4: Inference cost vs. r^2 value frontier across the pool (head sweep; encode cost is head-independent).

preparation at 19–20%, head training at 22–27%, model loading at about 7%, and evaluation below 0.1%. Feature extraction (model loading, data preparation, and inference) accounts for 73–78%.

Cost vs. value across the pool. Backbone encode cost spreads about 9× across the model pool (`potion-base-8M` at about 30 s vs. `mxbai-embed-large-v1` at about 256 s for the same data), while MLP-proxy quality varies substantially less. Figure 4 plots the resulting numbers. The static `model2vec` models lead the field. This is precisely what makes the pool-heterogeneity trap of Section 4.6 painful: the value-optimal frozen candidate is the one that cannot be fine-tuned.

5 CONCLUSION

We adapted feature-based model search to regression over tabular data serialized to text and evaluated it on a 16-encoder pool across two datasets. The experiments in Section 4 answer the two research questions as follows.

5.1 RQ1: Adaptation and Proxy Faithfulness

Adaptation succeeds, but it required three regression-specific changes, none of them "swap accuracy for r^2".

Our first adaptation is the replacement of the linear probe with a small MLP. The linear probe proxy, working for classification purposes, fails for regression (negative r^2, six times the spread, slowest training), while one hidden layer of 128–256 units fixes all of it. Second adaptation is the stabilization of the reference signal. LP-FT warmup cuts the cold-start divergences, target z-scoring removes all of them including the warmup-resistant `deberta-v3-base`, and gradient clipping and learning-rate decay are redundant on top. Obtaining a trustworthy reference ranking is itself part of the adaptation problem. Third, the quality metric also has to be adapted. On the long-tailed housing target, r^2 collapses the whole pool to zero while the models demonstrably learn the ordering (per-model prediction Spearman up to 0.78), and the metric and target-transform choice needs to be considered on a per-task level.

The seed-repeated measurement resolves what a single run cannot. The reference ranking is reproducible (seed-pair $\rho = 0.93$, ICC = 0.74 on the trainable band), so regret is a meaningful measurement. The frozen proxy still cannot fully reproduce the reference ranking ($\rho = 0.20$ $[0.07, 0.30]$), meaning the failure is the proxy's, not the reference's.

Yet the proxy is not useless. Its regret is near zero from budget 2 and zero from 3, and it beats the random-ranking null everywhere except $B = 1$, where it is worse than random for a systematic reason. It ranks the non-finetunable static embedders top, whose reference is capped at its frozen score. One lesson for the design of search systems is to filter, or separately track, candidates that cannot be fine-tuned before trusting the proxy ranking. With that filter, even regret@1 beats the null (0.006 vs. 0.018). The proxy's job on this task degrades from finding the best model to avoiding the bad ones, and it does the latter reliably well. The tight reference band of 0.046 r^2 for the primary dataset, about six times the noise floor, means that top-models perform similarly and it is irrelevant which of them is selected. The secondary dataset was meant to differentiate more between the well-performing models and instead exposed the target fragility.

5.2 RQ2: Bottlenecks

Alsatian's [25] outcomes agree with the regression transfer. Feature extraction, that is model loading, data preparation, and inference, takes 73–78% of proxy wall-clock, with backbone inference alone at 47–51%. Head training stays a secondary cost at 22–27%. The optimizations that classification systems target, such as embedding caching, search-order planning, and partial model access, remain the right levers for regression search.

The proxy/reference cost gap is the economic argument for using a proxy. Under our final configuration the reference sweep costs 15× the proxy sweep, 6–23× per trainable model, and up to 10× the peak GPU memory. The cheap pass serves a shortlist with good models at the top at roughly a fifteenth of the cost, provided the pool is filtered for fine-tunability. A regression-focused search system should invest in (i) MLP proxy heads, (ii) reference-pass stabilization through warmup and target standardization, (iii) a fine-tunability filter on the pool, (iv) the same feature-extraction optimizations as the classification literature, and (v) task-specific target transformations or adaptations.

6 OUTLOOK AND FUTURE WORK

Four extensions of the regression adaptation study follow directly from the findings.

Fixing the second dataset's metric. Rerunning `usa_real_estate` with a log-price or clipped target under the identical protocol is the one change that could let the pool separate and give the faithfulness verdict a second task family (Section 4.7). Our first experiments in that direction yielded promising results; however, due to a lack of time we let our verdict rest on the primary dataset alone.

Re-evaluating the research questions under the tuned configuration. The activation sweep and its full-evaluation follow-up were only conducted after the primary RQ1/RQ2 results had already been produced with ReLU as the head activation. Since tanh measurably changes how closely the proxy ranking tracks the reference the examination should be rerun.

Regret protocol refinement. The $B = 1$ pathology of Section 4.6 argues for making the fine-tunability filter a first-class part of the regret definition, or for always reporting the filtered curve alongside the full-pool curve, so that a non-finetunable top pick is a design decision rather than a coincidence.

Search-system optimizations. Two search-side mechanisms from the classification literature are natural next steps: successive halving over the pool, as in SHiFT [20], and on-disk embedding caching, for which the pipeline currently only carries a stub flag.

6.1 Transfer Pilot: Text Summarization

A separate track explores whether the proxy-vs-reference regret framing transfers to a generative task. Text summarization poses a new obstacle: unlike regression, it has no single ground-truth quality metric, so the reference signal $t(m, D)$ itself becomes a design choice. Candidate metrics can focus on n-gram overlap such as ROUGE (longest-common-subsequence) [10]; or semantic similarity such as BERTScore (contextual-embedding similarity) [31]; or BLEURT (BERTScore fine-tuned on human ratings) [23]; and classic LLM-as-a-judge. Factual-consistency scores can also be a quality dimension. The selection of the candidate pool is also open. Self-hostable LLMs spanning roughly an order of magnitude in active-parameter cost: Qwen3-30B-A3B-Instruct-2507 [17], GLM-4.5-Air [4], and GPT-OSS-120B [15]. Non-LLMs include BART (Meta) [9], PEGASUS (Google) [30], T5 (Google) [18], FLAN-T5 (Google) [2], and ProphetNet (Microsoft) [16]. Candidate datasets include CNN/DailyMail [22], XSum [12], SAMSum [3], and Newsroom [5], where tasks vary in size and difficulty. The pilot is purely exploratory, has not been implemented and thus produced no regret numbers yet, so we limit this section to the framing question: whether a frozen encoder score can predict fine-tuned summarization quality at all, and which reference metric such a proxy should be validated against; questions future work will have to answer.

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A RAW BENCHMARK RESULTS

This appendix records the full per-model results behind the aggregates in Section 4, plus the analysis figures not embedded in the body. All numbers are test-split values; single-run experiments are marked as such in the captions.

A.1 Candidate Pool

Table 8 lists the full 16-encoder candidate pool described in Section 3.6.

Table 8: Candidate pool: 16 pretrained text encoders. The `e5-base-v2` encoder prepends the prefix `query:` to every input.

model	adapter	pooling	dim
`all-MiniLM-L6-v2`	sent. transformers	builtin	384
`all-mpnet-base-v2`	sent. transformers	builtin	768
`sentence-t5-base`	sent. transformers	builtin	768
`mxbai-embed-large-v1`	sent. transformers	builtin	1024
`potion-base-8M`	model2vec	builtin	256
`potion-base-32M`	model2vec	builtin	512
`deberta-v3-small`	transformers enc.	mean	768
`deberta-v3-base`	transformers enc.	mean	768
`modernbert-base`	transformers enc.	mean	768
`modernbert-embed-base`	transformers enc.	mean	768
`distilbert-base-uncased`	transformers enc.	mean	768
`roberta-base`	transformers enc.	mean	768
`electra-base-discriminator`	transformers enc.	mean	768
`albert-base-v2`	transformers enc.	mean	768
`e5-base-v2`	transformers enc.	mean	768
`bge-base-en-v1.5`	transformers enc.	CLS	768

A.2 Proxy Repeatability

Table 9 lists the full frozen-proxy r^2 for all 16 models with one head fit versus the 3-seed prediction ensemble (FCH), behind the summary in Section 4.1.

Table 9: Frozen-proxy r^2 per model: one head fit vs. 3-seed prediction ensemble (FCH head, `wine_reviews`). Spearman correlation between the two rankings: $\rho = 0.991$.

model	1 repeat	3-seed ens.	Δ
`all-MiniLM-L6-v2`	−0.243	−0.243	+0.001
`all-mpnet-base-v2`	0.116	0.117	+0.001
`potion-base-8M`	−0.114	−0.114	−0.000
`potion-base-32M`	0.088	0.088	−0.000
`deberta-v3-small`	0.618	0.620	+0.002
`modernbert-base`	0.631	0.630	−0.001
`distilbert-base-uncased`	0.654	0.653	−0.001
`roberta-base`	0.673	0.673	+0.000
`deberta-v3-base`	0.406	0.406	+0.000
`electra-base-discriminator`	0.661	0.661	−0.000
`albert-base-v2`	0.667	0.665	−0.002
`e5-base-v2`	0.681	0.682	+0.001
`bge-base-en-v1.5`	0.609	0.622	+0.012
`modernbert-embed-base`	0.629	0.631	+0.002
`sentence-t5-base`	0.651	0.651	+0.000
`mxbai-embed-large-v1`	0.664	0.660	−0.003

A.3 Head Sweep: Per-Model Results

Figure 5 shows the model × head r^2 heatmaps of the head sweep (Section 4.4) for both passes; Tables 11 and 12 give the underlying full matrices, and Table 10 the per-head reference aggregates (pre-z-scoring protocol, one run per model and head).

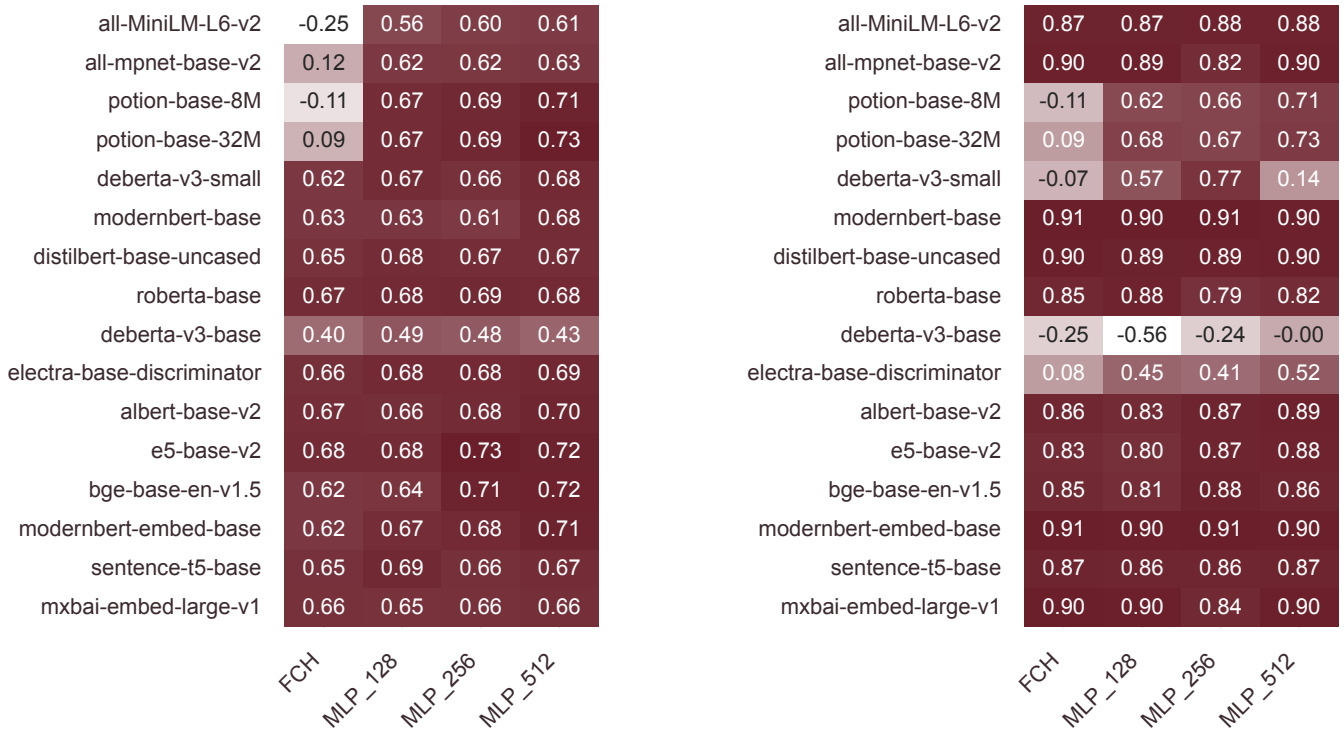


Figure 5: Model × head r^2 heatmaps for the head sweep: frozen proxy (left) and fine-tune reference (right).

Table 10: Fine-tune reference quality per head (16 models, one run each).

head	mean r^2	best model	best r^2	diverged
FCH	0.586	`modernbert-base`	0.911	2
`MLP_128`	0.706	`modernbert-base`	0.905	1
`MLP_256`	0.736	`modernbert-embed-base`	0.908	1
`MLP_512`	0.737	`modernbert-embed-base`	0.901	1

A.4 Warmup Ablation: Per-Model Results

Table 13 gives the per-model fine-tune r^2 for both arms of the warmup ablation. Figure 6 shows the corresponding heatmap and the divergence signature: diverged runs keep a high prediction-vs-target Spearman while r^2 goes negative.

A.5 Seed Statistics per Model

Table 14 gives the per-model seed statistics behind Section 4.6: mean and standard deviation of r^2 over the 5 seeds for both passes, and the range of ranks each model occupies. Figure 7 plots the same spread per model, and Figure 8 shows the seed × seed rank-agreement heatmaps.

A.6 Different Activation Functions

Tables 15 and 16 report the ReLU-vs-tanh comparison behind Section 4.5.

A.7 Dataset Comparison

Table 17 summarizes the four dataset configurations of Sections 4.7 and 4.8 under the fixed final recipe; Figure 9 shows the per-model r^2 strips.

A.8 Additional Figures

Figure 10 collects the remaining supplementary figures: the early-stopping behavior behind the epoch-cap numbers in Table 4, the head × head agreement of the proxy rankings, and the reference heatmap of the stabilizer 2×2.

Table 11: Head sweep: frozen-proxy r^2 per model and head.

model	FCH	MLP_128	MLP_256	MLP_512
all-MiniLM-L6-v2	−0.2484	0.5602	0.5997	0.6114
all-mpnet-base-v2	0.1159	0.6177	0.6237	0.6288
potion-base-8M	−0.1134	0.6738	0.6897	0.7104
potion-base-32M	0.0872	0.6711	0.6928	0.7331
deberta-v3-small	0.6210	0.6724	0.6573	0.6808
modernbert-base	0.6297	0.6281	0.6140	0.6762
distilbert-base-uncased	0.6485	0.6755	0.6669	0.6654
roberta-base	0.6726	0.6835	0.6887	0.6828
deberta-v3-base	0.4025	0.4902	0.4782	0.4300
electra-base-discriminator	0.6611	0.6806	0.6821	0.6938
albert-base-v2	0.6662	0.6628	0.6782	0.7003
e5-base-v2	0.6803	0.6788	0.7343	0.7180
bge-base-en-v1.5	0.6222	0.6377	0.7084	0.7223
modernbert-embed-base	0.6195	0.6685	0.6816	0.7107
sentence-t5-base	0.6502	0.6895	0.6578	0.6729
mxbai-embed-large-v1	0.6570	0.6496	0.6572	0.6615

Table 12: Head sweep: fine-tune reference r^2 per model and head. † marks fine-tunes flagged as diverged; $*$ marks the non-finetunable model2vec models, whose reference falls back to the frozen score. The isolated deberta-v3-small drop at MLP_512 (0.137) is single-run instability of this pre-z-scoring protocol, not a width effect.

model	FCH	MLP_128	MLP_256	MLP_512
all-MiniLM-L6-v2	0.8715	0.8721	0.8776	0.8778
all-mpnet-base-v2	0.8985	0.8936	0.8240	0.8985
potion-base-8M*	−0.1144	0.6250	0.6589	0.7062
potion-base-32M*	0.0862	0.6814	0.6656	0.7253
deberta-v3-small	−0.0682†	0.5696	0.7709	0.1368
modernbert-base	0.9106	0.9049	0.9057	0.8954
distilbert-base-uncased	0.9021	0.8864	0.8925	0.8985
roberta-base	0.8522	0.8799	0.7880	0.8223
deberta-v3-base	−0.2523†	−0.5632†	−0.2398†	−0.0000†
electra-base-discriminator	0.0845	0.4459	0.4079	0.5233
albert-base-v2	0.8556	0.8292	0.8662	0.8916
e5-base-v2	0.8300	0.7973	0.8655	0.8790
bge-base-en-v1.5	0.8465	0.8094	0.8823	0.8618
modernbert-embed-base	0.9058	0.9019	0.9081	0.9010
sentence-t5-base	0.8693	0.8604	0.8619	0.8673
mxbai-embed-large-v1	0.9049	0.9034	0.8448	0.9000

Table 13: Warmup ablation: fine-tune reference r^2 per model (`MLP_256` head). † marks diverged fine-tunes; ∗ marks the non-finetunable `model2vec` models.

model	warmup (2 ep.)	no warmup
all-MiniLM-L6-v2	0.8776	0.8702
all-mpnet-base-v2	0.8240	0.8924
potion-base-8M*	0.6589	0.6809
potion-base-32M*	0.6656	0.7242
deberta-v3-small	0.7709	$-0.2584^{\dagger}$
modernbert-base	0.9057	0.9000
distilbert-base-uncased	0.8925	0.7979
roberta-base	0.7880	$-0.1177^{\dagger}$
deberta-v3-base	$-0.2398^{\dagger}$	$-0.6227^{\dagger}$
electra-base-discriminator	0.4079	0.2049
albert-base-v2	0.8662	0.8731
e5-base-v2	0.8655	0.7032
bge-base-en-v1.5	0.8823	0.6187
modernbert-embed-base	0.9081	0.9129
sentence-t5-base	0.8619	0.8665
mxbai-embed-large-v1	0.8448	0.8048

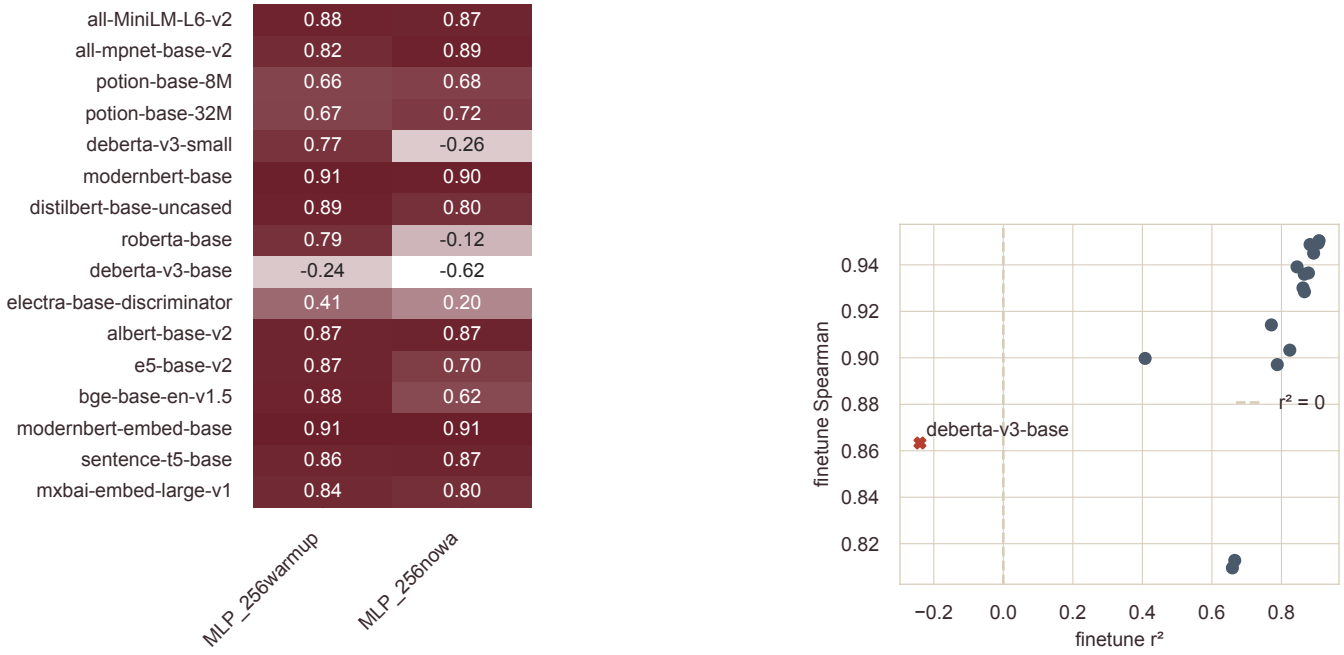


Figure 6: Warmup ablation: fine-tune reference r^2 heatmap (left) and the divergence signature (right): prediction-vs-target Spearman stays high while r^2 collapses, the ranking-preserving scale blow-up of Section 4.2.

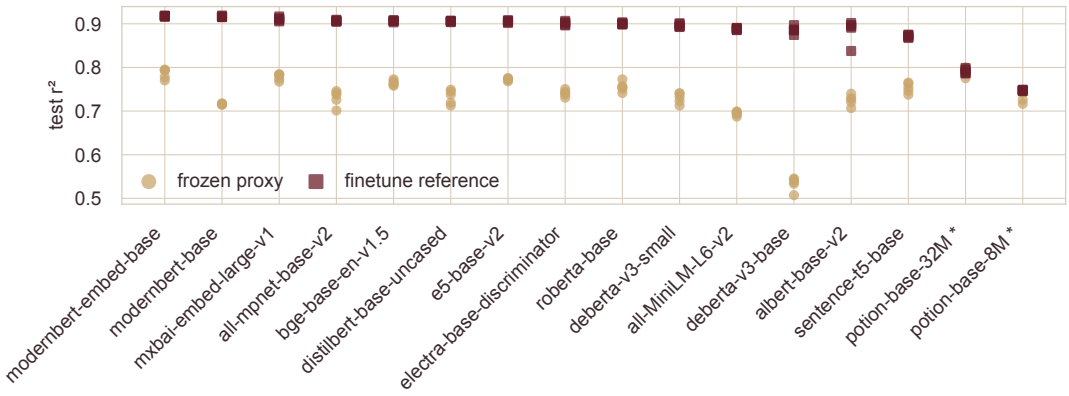


Figure 7: Per-model r^2 across the 5 seeds, frozen and fine-tune passes. The within-model spread (noise floor) is small against the between-model band.

Table 14: Per-model r^2 across 5 seeds under the final recipe (`wine_reviews`), sorted by fine-tune mean. $*$ marks the non-finetunable `model2vec` models.

	frozen		fine-tune		rank range	
model	**mean**	**std**	**mean**	**std**	**fz**	**ft**
`modernbert-embed-base`	0.786	0.011	0.918	0.001	1–3	1–2
`modernbert-base`	0.716	0.001	0.917	0.002	10–14	1–3
`mxbai-embed-large-v1`	0.779	0.007	0.911	0.004	2–4	2–7
`all-mpnet-base-v2`	0.730	0.018	0.907	0.002	9–14	4–7
`bge-base-en-v1.5`	0.765	0.006	0.906	0.002	4–6	4–8
`distilbert-base-uncased`	0.733	0.016	0.906	0.001	7–13	4–7
`e5-base-v2`	0.773	0.003	0.906	0.003	2–6	3–8
`electra-base-discriminator`	0.741	0.007	0.901	0.004	7–11	6–9
`roberta-base`	0.756	0.011	0.901	0.002	5–8	7–11
`deberta-v3-small`	0.730	0.012	0.896	0.004	10–13	10–11
`all-MiniLM-L6-v2`	0.694	0.005	0.888	0.002	15	11–13
`deberta-v3-base`	0.533	0.015	0.886	0.008	16	12–14
`albert-base-v2`	0.725	0.012	0.884	0.026	11–13	9–14
`sentence-t5-base`	0.753	0.012	0.872	0.003	5–11	13–14
`potion-base-32M`*	0.793	0.010	0.792	0.005	1–2	15
`potion-base-8M`*	0.736	0.014	0.747	0.001	8–12	16

Table 15: Frozen pass (proxy) — ReLU vs. tanh, `MLP_256`, `wine_reviews`, 16-model pool, target z-scoring on.

metric	**ReLU (job 2333233)**	**tanh (job 2375093)**
mean proxy r^2 (16 models)	0.735	0.706
std proxy r^2 (16 models)	0.062	0.078
mean proxy r^2 (14, excl. model2vec)	0.731	0.706
proxy top-1 model	`modernbert-embed-base` (0.791)	`modernbert-embed-base` (0.786)
min proxy r^2 (16 models)	`deberta-v3-base` (0.517)	`deberta-v3-base` (0.527)
proxy-vs-ref Spearman (16 models)	0.385	0.662
proxy-vs-ref Spearman (14, excl. model2vec)	0.596	0.692

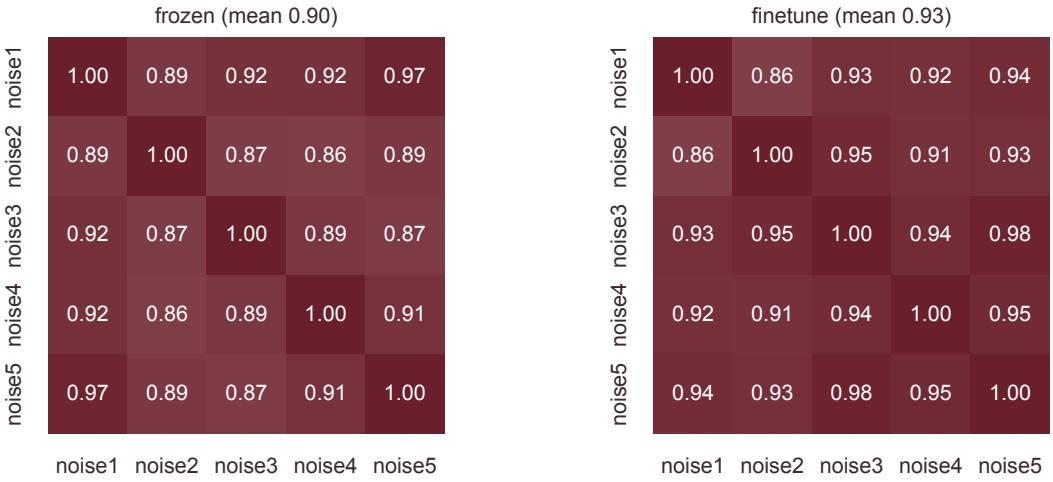


Figure 8: Seed × seed Spearman rank agreement of the frozen (left) and fine-tune (right) rankings under the final recipe.

Table 16: Fine-tune pass (reference) — ReLU vs. tanh, `MLP_256`, `wine_reviews`, 16-model pool, target z-scoring on. ∗ marks the non-finetunable `model2vec` model driving the min.

metric	ReLU (job 2333233)	tanh (job 2375093)
mean ref r^2 (16 models)	0.881	0.865
mean ref r^2 (14, excl. model2vec)	0.898	0.899
ref top-1 model	`modernbert-embed-base` (0.917)	`modernbert-embed-base` (0.917)
min ref r^2 (16 models)	`potion-base-8M`* (0.747)	`potion-base-8M`* (0.605)
regret@1	0.000	0.000

Table 17: Per-dataset summary under the final recipe (16 models each). Prediction ρ is the mean per-model prediction-vs-target Spearman after fine-tuning. The single flat housing run counted as diverged is not a wine-style scale blow-up, and housing's low regret@1 carries no signal: the pool does not separate there, so any ranking incurs near-zero regret (Section 4.7).

dataset	frozen r^2	fine-tune r^2	pred. ρ	div.	regret@1
wine full	0.53–0.79	0.75–0.92	0.95	0	0.100
wine tiny	0.32–0.68	0.39–0.74	0.83	0	0.012
housing full	−0.00–0.02	−0.00–0.03	0.65	1	0.013
housing tiny	0.04–0.14	0.01–0.26	0.63	0	0.047

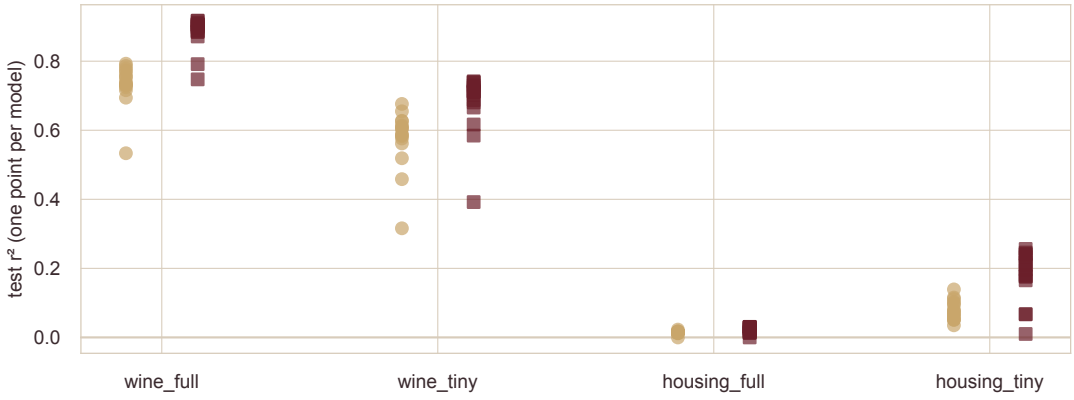


Figure 9: Per-model r^2 strips across wine and housing, full and tiny data budgets, both passes.

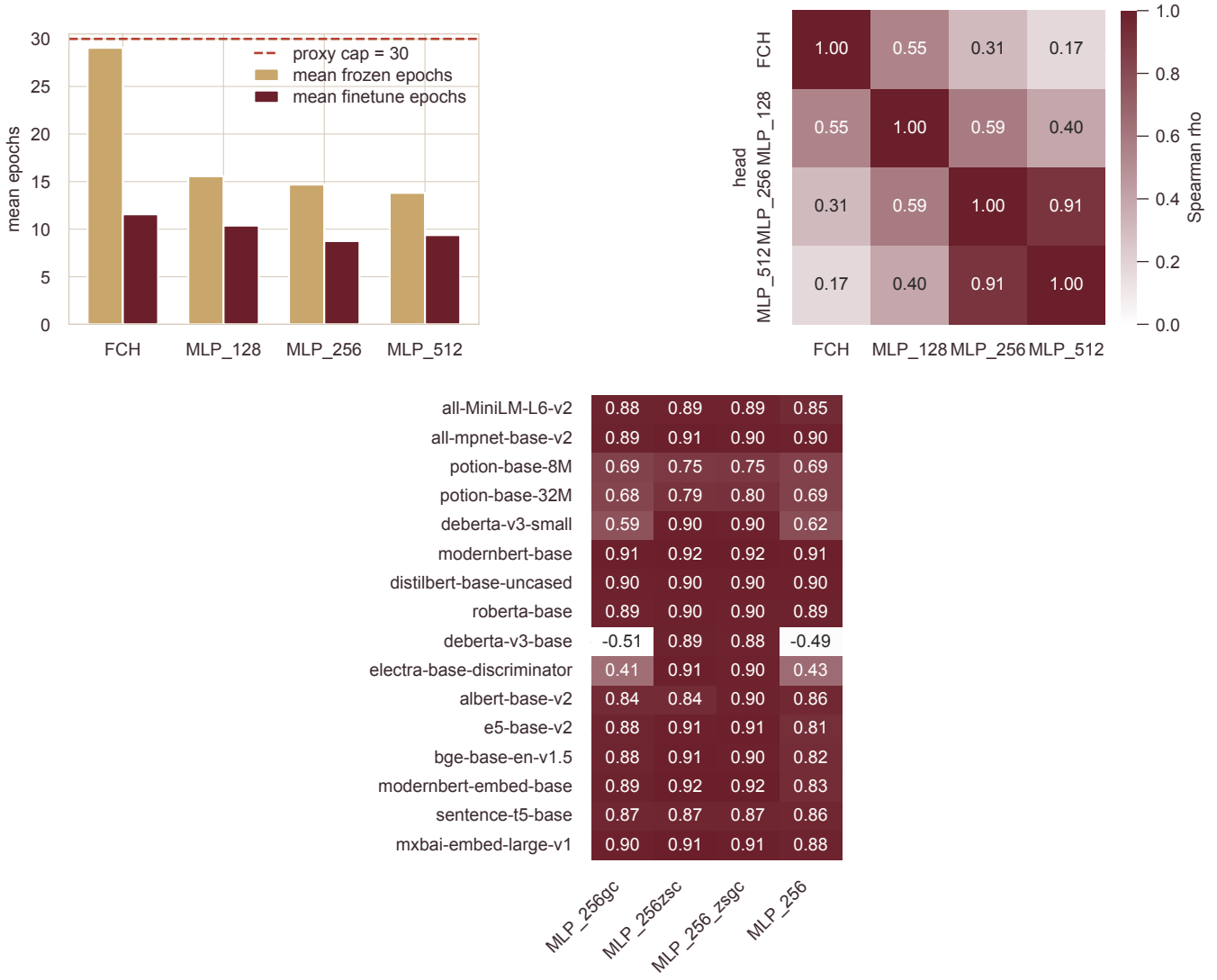


Figure 10: Supplementary figures. Top left: mean epochs run per head and pass (the FCH proxy sits at the 30-epoch cap for 12 of 16 models). Top right: Spearman rank agreement between the frozen-proxy rankings of different heads. Bottom: fine-tune reference r^2 heatmap of the stabilizer 2×2, whose columns run `MLP_256gc` (gradient clipping only), `MLP_256zsc` (target z-scoring only), `MLP_256_zsgc` (both), and `MLP_256` (the warmup-2 baseline with neither).



Figure 11: Per-arm regret@1 for the stabilizer 2×2, companion to Table 3 in Section 4.3. Reading the axis left to right, `MLP_256gc` is gradient clipping only, `MLP_256zsc` target z-scoring only, `MLP_256_zsgc` both, and `MLP_256` the warmup-2 baseline with neither stabilizer.